9.6.5 Pole pitch

It is the distance between the center of two adjacent poles. It is also denoted by the number of slots per pole.

$$\text{Pole pitch} = \frac{number\ of\ slots}{number\ of\ poles}$$

9.6.6 Winding pitch or coil pitch or coil span

It is the distance between the two coil sides of a coil.

Coil pitch = 1+ pole pitch

9.6.7 Full pitched winding

In full pitched winding, pole pitch and coil pitch are equal. There will be 180° (electrical degree) between the coil sides.

9.6.8 Fractional pitched winding

If we reduce or increase the coil pitch by one or two in number than the pole pitch, then it is called as fractional pitched winding. The angle (electrical degree) between the coil sides may be less than or more than 180°.

Fractional pitched winding may be divided into two types. They are

i. Short pitched winding (or) short chorded winding
ii. Long chorded winding

9.6.9 Pitch factor (K_c)

It is the ratio between the coil pitch and pole pitch. It is denoted by 'K_c'.

$$\text{Pitch factor, } K_c = \frac{coil\ pitch}{pole\ pitch} = cos\frac{\alpha}{2}$$

9.6.10 Short pitched winding or short chorded winding

If we reduce the coil pitch by one or two in number than the pole pitch, then it is called as short pitched winding. The reducing value should not exceed 1/3 rd of the pole pitch value. Electrical degree between two coil sides of short pitched winding will be less than 180°. This reduced angle is represented by short pitch angle (α). Here pitch factor will be less than one. This process reduces the weight of copper needed for winding the motors.

9.6.11 Long chorded winding

If we extend the coil pitch by one or two than the pole pitch, it is called as long chorded winding. Electrical degree between two coil sides of long chorded winding will be more than 180°. Here pitch factor will be more than one. This method of winding is rarely used.

9.6.12 Single layer winding

If each slot of a motor contains one coil side, then it is called single layer winding. Here the number of coils will be equal to the half of the total number of slots.

Hence,

$$\text{Number of coils} = \frac{\text{Total number of slots.}}{2}$$

9.6.13 Double layer winding

In double layer winding, two different coil sides of two coils are inserted in the same slot. Here the number of coils will be equal to the total number of slots.

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